Substitution of Zn^{2+} by Al^{3+} and Ga^{3+} in Hydrozincite – Substitution or formation of side phases?

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Introduction

Hydrozincite ($Zn_5(OH)_6(CO_3)_2$) is a naturally occurring mineral with a layered structure which can be prepared synthetically by co-precipitation in large quantities. [1] It is a suitable precursor for the preparation of ZnO by thermal decomposition e.g. for the use as support material for catalysts. [2] Further modification by incorporation of e.g. Al^{3+} or Ga^{3+} is desirable, since these secondary promotors enhance the activity of Cu/ZnO based catalysts for methanol synthesis. [3,4] For maximizing the dispersion of Al^{3+}/Ga^{3+} in ZnO the formation of a single-source precursor is mandatory and segregation in a side phase must be avoided.

Results

The samples were prepared by co-precipitation in an automated reactor system. For the sample containing solely Zn^{2+} the X-ray powder diffraction (XRPD) data evidences phase pure hydrozincite (Fig. 1). With increasing Ga^{3+} content two key observations can be made: First, the reflections of this phase broaden anisotropically. This indicates both smaller crystallites but also some disorder in the stacking pattern of the layers.

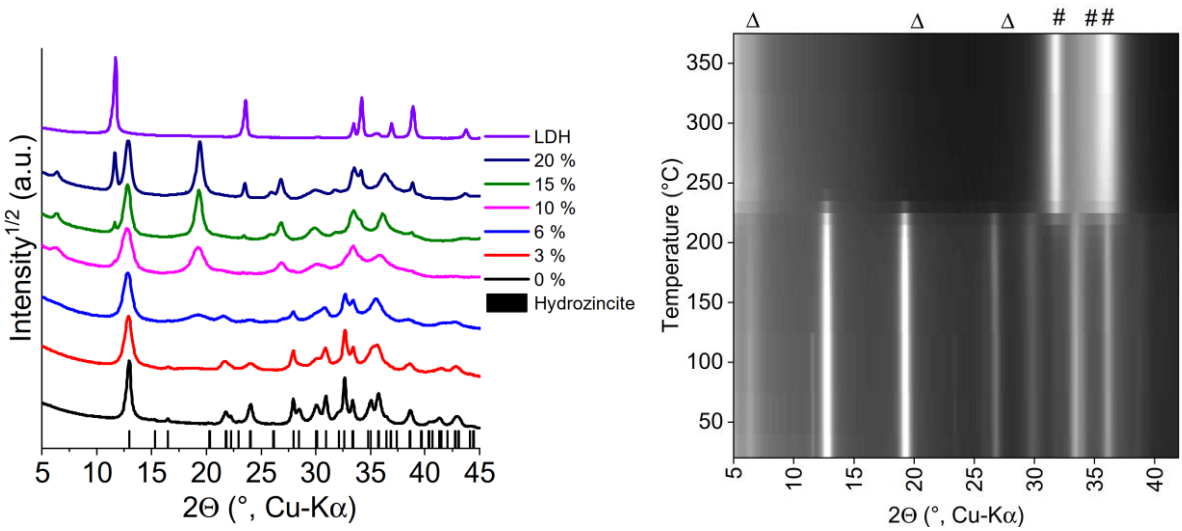

Fig. 1: Left: XRPD data of the hydrozincites containing different amounts of Ga^{3+} as indicated (metals basis). Right: Temperature resolved XRPD data for the hydrozincite containing 15 at.-% Al^{3+}.

Secondly, additional reflections appear. The straight forward interpretation would be the formation of a second crystalline phase. However, in the temperature resolved XRPD data of the sample containing 15 at.-% Ga^{3+} all of the reflections vanish simultaneously around 230 °C, which contradicts a side phase where a different decomposition temperature is expected. The position of the first two additional reflections at 6.5 and 19° 2Θ can be explained assuming a lower symmetry, e.g. when lowering from *C*2/*m* to *P*2/*m*.

A key question is what structural changes occur in the lower symmetry hydrozincite. Since the broadening of the reflections in the XRPD pattern hampers a straight forward analysis, we turned to the pair distribution function (PDF) to analyze changes in the materials local structure. The data (Fig. 2) shows gradual changes with chemical composition. Already at low interatomic distances substantial changes are evident: The layers of hydrozincite contain Zn^{2+} in both octahedral and tetrahedral coordination. The distance between adjacent octahedral sites is slightly shorter than the distance between an octahedral and a tetrahedral or two tetrahedral sites. Those correspond to the two signals between 3-4 Å in the PDF. The latter decreases largely in intensity, which points to a deoccupation of tetrahedral sites upon substitution of Zn^{2+} by Ga^{3+}. Further details as well as similarities and differences observed for the case of Al^{3+} will be presented. To summarize, the results so far strongly indicate that hydrozincite is capable to incorporate trivalent cations by substitution, undergoing significant structural changes.

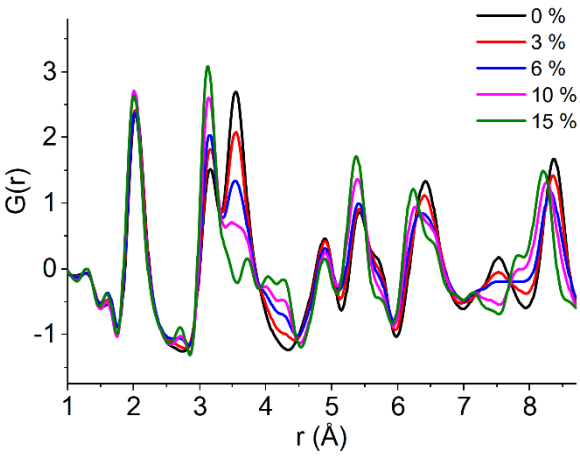

Fig. 2: PDF data of the hydrozincites containing different amounts of Ga^{3+}.

References

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